

Predicting Factor Decay: ML Models for Cross-Factor Predictability Erosion

CHOROK LEE / KAIST

Original ten-page TeX preprint edition; separate from the five-page submission.

Forecasting the weakening of factor relationships remains exploratory

RESEARCH QUESTION

Can the past behavior of a predictive relationship indicate when it will lose statistical significance?

30

directed factor pairs

7

decay events

6/10

correct holdout classifications

Motivation

A cross-factor signal can look useful in one period and weak in another. This study asks whether that weakening can itself be modeled. The main example uses past value-factor returns (HML) to predict the size factor (SMB). Granger predictability describes incremental information within a specified model.

Decay model

Features describe initial F strength, rolling signal behavior, correlation, and relative volatility. The manuscript reports strong cross-validated ranking, but only seven decay events underpin the analysis. A separate test on ten factor pairs yields six correct classifications. Pair splits do not establish performance on future dates. The displayed variation is a fold standard deviation, not a confidence interval.

Operational distinction

The rolling HMM detects regime changes with a reported mean delay of 6.6 days, accuracy of 47.6%, and F1 of 0.39. This measures detection after a regime transition. It does not represent advance warning of future factor decay.

Study design

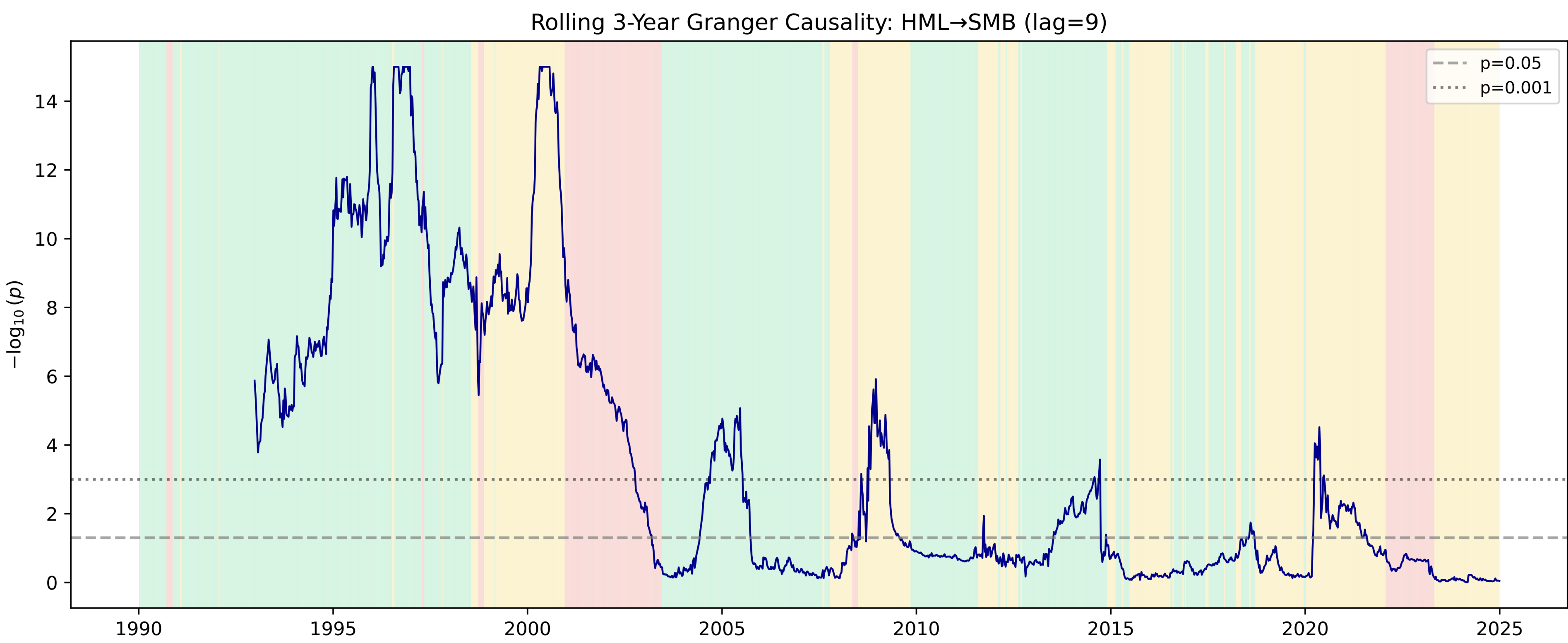
The primary panel contains six daily equity factors over 1990-2024. Student-t hidden Markov models identify statistical regimes. Granger tests characterize 30 directed factor pairs. Classifiers predict a loss of significance within five years, while survival models model the timing of a decay event.

Later-period evidence

For the main US value-to-size relationship, the 2013-2024 evaluation fails corrected significance tests. Selected crisis-state results in Developed ex-US and Asia-Pacific ex-Japan survive the reported regional correction. These relationship-level results do not independently validate the survival predictor. The source notes that return scaling changes regime assignments.

EVIDENCE AT A GLANCE

Evaluation	Reported metric (± fold SD)
Survival forest, 5-fold CV	C-index 0.933 ± 0.097
Logistic model, 5-fold CV	AUC 0.933 ± 0.133
Ten-pair holdout	Accuracy 60%



Original rolling Granger figure. Vertical axis: -log10(p). Three-year windows, source lags 1-9. Evidence of predictability varies through time.

TAKEAWAY

The framework motivates monitoring the durability of predictive relationships. The current evidence is exploratory and does not establish a profitable trading rule or a structural causal mechanism.

INTERPRETATION & LIMITS

Only seven events; shared factors across pairs; post-hoc selection; broad specification search. Some decay fits use full-sample regime assignments. The manuscript gives inconsistent state-selection windows. Fully chronological refitting remains necessary.